



运筹学基础及其 Matlab 应用

实验项目名称 _____

所属课程名称 _____

作业日期 _____

姓名（学号） _____

成绩 _____

2. 利用单纯形法求解下面的问题。

$$\begin{aligned}
 (1) \quad & \min \quad z = P_1 d_1^- + P_2 (2d_2^+ + d_3^-) \\
 & s.t. \quad \begin{cases} 8x_1 + 4x_2 + d_1^- - d_1^+ = 160 \\ x_1 + 2x_2 + d_2^- - d_2^+ = 30 \\ x_1 + 2x_2 + d_3^- - d_3^+ = 40 \\ x_1, x_2 \geq 0, d_i^-, d_i^+ \geq 0, i = 1, 2, 3 \end{cases} \\
 (3) \quad & \min \quad z = P_1 (d_1^- + d_1^+) + P_2 (d_2^- + d_3^+) \\
 & s.t. \quad \begin{cases} x_1 + x_2 + d_1^- - d_1^+ = 40 \\ 2x_1 + x_2 + d_2^- - d_2^+ = 14 \\ x_2 + d_3^- - d_3^+ = 30 \\ x_1, x_2 \geq 0, d_i^-, d_i^+ \geq 0, i = 1, 2, 3 \end{cases}
 \end{aligned}$$

(1)

表 4.1 习题2(1)初始单纯形表及其迭代

c_j			0	0	P_1	0	0	$2P_2$	P_2	0	θ
c_B	X_B	b	x_1	x_2	d_1^-	d_1^+	d_2^-	d_2^+	d_3^-	d_3^+	
P_1	d_1^-	160	[8]	4	1	-1	0	0	0	0	20
0	d_2^-	30	1	2	0	0	1	-1	0	0	30
P_2	d_3^-	40	1	2	0	0	0	0	1	-1	40
$\sigma_j = c_j - z_j$		P_1	-8	-4	0	1	0	0	0	0	
		P_2	-1	-2	0	0	0	2	0	1	
0	x_1	20	1	1/2	1/8	-1/8	0	0	0	0	40
0	d_2^-	10	0	[3/2]	-1/8	1/8	1	-1	0	0	20/3
P_2	d_3^-	20	0	3/2	-1/8	1/8	0	0	1	-1	40/3
$\sigma_j = c_j - z_j$		P_1	0	0	1	0	0	0	0	0	
		P_2	0	-3/2	1/8	-1/8	0	2	0	1	
0	x_1	50/3	1	0	1/6	-1/6	-1/3	1/3	0	0	
0	x_2	20/3	0	1	-1/12	1/12	2/3	-2/3	0	0	
P_2	d_3^-	10	0	0	0	0	-1	1	1	-1	
$\sigma_j = c_j - z_j$		P_1	0	0	1	0	0	0	0	0	
		P_2	0	0	0	0	1	1	0	1	

故得到满意解为 $x_1 = \frac{50}{3}, x_2 = \frac{20}{3}, d_3^- = 10$, 此时满意值为 $z = 10P_2$ 。

(3)

c_j			0	0	P_1	P_1	P_2	0	0	P_2	θ
c_B	X_B	b	x_1	x_2	d_1^-	d_1^+	d_2^-	d_2^+	d_3^-	d_3^+	
P_1	d_1^-	40	1	1	1	-1	0	0	0	0	40
P_2	d_2^-	14	[2]	1	0	0	1	-1	0	0	7
0	d_3^-	30	0	1	0	0	0	0	1	-1	-
$\sigma_j(P_1)$			-1	-1	0	2	0	0	0	0	
$\sigma_j(P_2)$			-2	-1	0	0	0	1	0	1	
P_1	d_1^-	33	0	$\frac{1}{2}$	1	-1	$-\frac{1}{2}$	$\frac{1}{2}$	0	0	66
0	x_1	7	1	$[\frac{1}{2}]$	0	0	$\frac{1}{2}$	$-\frac{1}{2}$	0	0	14
0	d_3^-	30	0	1	0	0	0	0	1	-1	30
$\sigma_j(P_1)$			0	$-\frac{1}{2}$	0	2	$\frac{1}{2}$	$-\frac{1}{2}$	0	0	
$\sigma_j(P_2)$			0	0	0	0	1	0	0	1	
P_1	d_1^-	26	-1	0	1	-1	-1	1	0	0	26
0	x_2	14	2	1	0	0	1	-1	0	0	-
0	d_3^-	16	-2	0	0	0	-1	[1]	1	-1	16
$\sigma_j(P_1)$			1	0	0	2	1	-1	0	0	
$\sigma_j(P_2)$			0	0	0	0	1	0	0	1	
P_1	d_1^-	10	[1]	0	1	-1	0	0	-1	1	10
0	x_2	30	0	1	0	0	0	0	1	-1	-
0	d_2^+	16	-2	0	0	0	-1	1	1	-1	-
$\sigma_j(P_1)$			-1	0	0	2	0	0	1	-1	
$\sigma_j(P_2)$			0	0	0	0	1	0	0	1	
0	x_1	10	1	0	1	-1	0	0	-1	1	10
0	x_2	30	0	1	0	0	0	0	1	-1	-
0	d_2^+	36	0	0	2	-2	-1	1	-1	1	-
$\sigma_j(P_1)$			0	0	1	1	0	0	0	0	
$\sigma_j(P_2)$			0	0	0	0	1	0	0	1	

故得到满意解为 $x_1 = 10, x_2 = 30, d_2^+ = 36$, 此时满意值为 $z = 0$ 。

3 考虑如下目标规划问题:

$$\begin{aligned} \min \quad & z = P_1(2d_1^+ + 3d_2^+) + P_2d_3^- + P_3d_4^- \\ \text{s.t.} \quad & \end{aligned}$$

$$\begin{cases} x_1 + x_2 + d_1^- - d_1^+ = 10 \\ x_1 + d_2^- - d_2^+ = 4 \\ 5x_1 + 3x_2 + d_3^- - d_3^+ = 56 \\ x_1 + x_2 + d_4^- - d_4^+ = 12x_1, x_2 \geq 0, d_i^-, d_i^+ \geq 0, i = 1, 2, 3, 4. \end{cases}$$

(1)用单纯形法求此目标规划问题; (2)分析当第二和第三优先目标交换后满意解的变化。

c_j	0	0	0	$2P_1$	0	$3P_1$	P_2	0	P_3	0	θ
c_B	x_1	x_2	d_1^-	d_1^+	d_2^-	d_2^+	d_3^-	d_3^+	d_4^-	d_4^+	
0	d_1^-	10	1	1	1	-1	0	0	0	0	10
0	d_2^-	4	$[1]$	0	0	0	1	-1	0	0	4
P_2	d_3^-	56	5	3	0	0	0	0	1	-1	$\frac{56}{5}$
P_3	d_4^-	12	1	1	0	0	0	0	0	1	12
$\sigma_j(P_1)$			0	0	0	2	0	3	0	0	
$\sigma_j(P_2)$			-5	-3	0	0	0	0	1	0	
$\sigma_j(P_3)$			-1	-1	0	0	0	0	0	1	
0	d_1^-	6	0	$[1]$	1	-1	-1	1	0	0	6
0	x_1	4	1	0	0	0	1	-1	0	0	$-$
P_2	d_3^-	36	0	3	0	0	-5	5	1	-1	12
P_3	d_4^-	8	0	1	0	0	-1	1	0	1	8
$\sigma_j(P_1)$			0	0	0	2	0	3	0	0	
$\sigma_j(P_2)$			0	-3	0	0	5	-5	0	1	
$\sigma_j(P_3)$			0	-1	0	0	1	-1	0	0	
0	x_2	6	0	1	1	-1	-1	1	0	0	
0	x_1	4	1	0	0	0	1	-1	0	0	
P_2	d_3^-	18	0	0	-3	3	-2	2	1	-1	
P_3	d_4^-	2	0	0	-1	1	0	0	0	1	
$\sigma_j(P_1)$			0	0	0	2	0	3	0	0	
$\sigma_j(P_2)$			0	0	3	-3	2	-2	0	1	
$\sigma_j(P_3)$			0	0	1	-1	0	0	0	1	

故得到满意解为 $x_1 = 4, x_2 = 6, d_3^- = 18, d_4^- = 2$, 此时满意值为 $18P_2 + 2P_3$ 。

(2) 由下方单纯形表可知, 交换目标后满意解变为 $2P_2 + 18P_3$ 。

c_j	0	0	0	$2P_1$	0	$3P_1$	P_3	0	P_2	0	θ
c_B	x_1	x_2	d_1^-	d_1^+	d_2^-	d_2^+	d_3^-	d_3^+	d_4^-	d_4^+	
0	d_1^-	10	1	1	1	-1	0	0	0	0	10
0	d_2^-	4	$[1]$	0	0	0	1	-1	0	0	4
P_3	d_3^-	56	5	3	0	0	0	0	1	-1	$\frac{56}{5}$
P_2	d_4^-	12	1	1	0	0	0	0	0	1	12
$\sigma_j(P_1)$			0	0	0	2	0	3	0	0	
$\sigma_j(P_2)$			-1	-1	0	0	0	0	0	1	
$\sigma_j(P_3)$			-5	-3	0	0	0	0	1	0	
0	d_1^-	6	0	$[1]$	1	-1	-1	1	0	0	6
0	x_1	4	1	0	0	0	1	-1	0	0	$-$
P_3	d_3^-	36	0	3	0	0	-5	5	1	-1	12
P_2	d_4^-	8	0	1	0	0	-1	1	0	1	8
$\sigma_j(P_1)$			0	0	0	2	0	3	0	0	
$\sigma_j(P_2)$			0	-1	0	0	1	-1	0	0	
$\sigma_j(P_3)$			0	-3	0	0	5	-5	0	1	
0	x_2	6	0	1	1	-1	-1	1	0	0	
0	x_1	4	1	0	0	0	1	-1	0	0	
P_3	d_3^-	18	0	0	-3	3	-2	2	1	-1	
P_2	d_4^-	2	0	0	-1	1	0	0	0	1	
$\sigma_j(P_1)$			0	0	0	2	0	3	0	0	
$\sigma_j(P_2)$			0	0	1	-1	0	0	0	1	
$\sigma_j(P_3)$			0	0	3	-3	2	-2	0	1	

4 已知某线性规划问题为

$$\begin{aligned} \max \quad & z = 100x_1 + 50x_2 \\ \text{s.t.} \quad & \begin{cases} 10x_1 + 16x_2 \leq 200 \text{ (资源1)} \\ 11x_1 + 3x_2 \geq 25 \text{ (资源2)} \\ x_1, x_2 \geq 0 \end{cases} \end{aligned}$$

假设重新确定该问题的目标为： P_1 ，目标函数 z 的值不低于1900； P_2 ，资源1必须全部用完。在此假设下将其转换为目标规划问题并列出数学模型然后求其满意解。

解 根据题目要求可建立如下目标规划模型：

$$\begin{aligned} \min \quad & z = P_1d_1^- + P_2(d_2^- + d_2^+) \\ \text{s.t.} \quad & \begin{cases} 100x_1 + 50x_2 + d_1^- - d_1^+ = 1900 \\ 10x_1 + 16x_2 + d_2^- - d_2^+ = 200 \\ 11x_1 + 3x_2 \geq 25 \\ x_1, x_2 \geq 0, d_i^-, d_i^+ \geq 0, i = 1, 2. \end{cases} \end{aligned}$$

在上述模型中的第三个约束条件中引进剩余变量 $x_s \geq 0$ ，并引进人工变量 $x_3 \geq 0$ ，则有如下标准形式：

$$\begin{aligned} \min \quad & z = P_1d_1^- + P_2d_2^- + P_2d_2^+ + Mx_3 \\ \text{s.t.} \quad & \begin{cases} 100x_1 + 50x_2 + d_1^- - d_1^+ = 1900 \\ 10x_1 + 16x_2 + d_2^- - d_2^+ = 200 \\ 11x_1 + 3x_2 - x_s + x_3 = 25 \\ x_1, x_2 \geq 0, d_i^-, d_i^+ \geq 0, i = 1, 2. \end{cases} \end{aligned}$$

然后利用单纯形法之大 M 法求解此问题，见表(4.6)。从表中得知原问题的满意解为 $x_1^* = 204/11 = 18.5455, x_2^* = 10/11 = 0.9091, d_1^- = 0, d_1^+ = 0, d_2^- = 0, d_2^+ = 0, x_s = 1999/11 = 181.7273$ ，容易验证，在达到第一优先目标的基础上也完成了第二优先目标。如果直接求解原来的问题，即不特别确定优先目标，直接求解原来的极大化问题（过程从略）则有 $x_1^* = 20, x_2^* = 0, z^* = 2000$ 。显然，此时也完成了目标函数值不低于1900的要求，且资源1也全部用完了。

表 4.6 习题4初始单纯形表及最终单纯形表

c_j			0	0	P_1	0	P_2	P_2	0	M	θ
c_B	X_B	b	x_1	x_2	d_1^-	d_1^+	d_2^-	d_2^+	x_s	x_3	
P_1	d_1^-	1900	100	50	1	-1	0	0	0	0	19
P_2	d_2^-	200	10	16	0	0	1	-1	0	0	20
M	x_3	25	[11]	3	0	0	0	0	-1	1	25/11
$\sigma_j = c_j - z_j$			P_1	-100	-50	0	1	0	0	0	
			P_2	-10	-16	0	0	0	2	0	
			M	-11	-3	0	0	0	0	1	
0	x_1	204/11	1	0	4/275	-4/275	-1/22	1/22	0		
0	x_s	1999/11	0	0	73/550	-73/550	-5/22	5/22	1		
0	x_2	10/11	0	1	-1/110	1/110	1/11	-1/11	0		
$\sigma_j = c_j - z_j$			P_1	0	0	1	0	0	0	0	
			P_2	0	0	0	0	1	1	0	
			M	0	0	0	0	0	0	0	

